

AWS Cloud Migration Strategy for NorthStar Digital Health

Cloud Computing Module

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1 Introduction

Cloud computing has become a very significant option for companies that need scalability of their infrastructure, global reach, and cost management flexibility. Instead of relying only on physical hardware and internal data centers, cloud computing allows organizations to utilize computing assets like processing power, storage space, databases, networks, and security from the internet. One of the most notable examples of cloud computing is Amazon Web Services, which is often employed by companies to update their infrastructure and get around the constraints of conventional solutions.

NorthStar Digital Health (NDH) is a mid-sized company that builds software for the healthcare industry and provides web and mobile applications for doctors, insurance companies, and wellness institutions around the world in Europe, North America, and the Asia-Pacific region. The platforms developed by NDH include patient engagement capabilities, scheduling appointments, analytics dashboards, and machine learning technologies for process optimization and medical image analysis. This kind of service places high demands on performance, reliability, security, and regulatory compliance due to the nature of the NDH business area.

Currently, NDH has significant on-premises infrastructural resources including code storage, internal networks, development and testing databases, application servers, production servers, and GPUs for artificial intelligence tasks. However, scaling, maintaining, and using such on-premises infrastructural resources have become increasingly challenging, expensive, and inappropriate given the growing international labor force and customers of the company. The research proposes a strategy for cloud migration to AWS for NDH. Migration planning, application refactoring, scalable architecture, cost savings, global expansion, security, compliance, and operations management are all considered within the research context.

2 Organizational Context and Migration Drivers

2.1 NorthStar Digital Health Overview

The business model of NDH is centered on the constant provision of secure, reliable, and highly-performing digital services. In contrast to companies that use their IT infrastructure only for their own administrative processes, the IT infrastructure of NDH is an essential component of its services offered to the healthcare industry. Therefore, decisions on infrastructure directly affect the quality of services, customer satisfaction, and opportunities for expansion.

2.2 Current On-Premise Infrastructure

The NDH on-premise environment currently has some key technological elements, which include code repositories, internal network, databases for development and testing, software development servers, servers for running applications, and GPUs for running AI and data-heavy operations. The above technologies support both internal development processes and external healthcare operations conducted by NDH.

However, being on premise provides control of the technological infrastructure but comes with limitations in terms of flexibility. Servers need to be bought, configured, upgraded, and eventually decommissioned. Thus, the planning process for these operations becomes long and makes adapting to changes difficult since if an increased capacity is needed, the company has to purchase the technology and deploy it before using it to conduct development or other operations.

3 Migration Assessment and Strategy

3.1 Workload Assessment

Prior to moving their infrastructure to AWS, NDH must conduct a detailed review of its existing on-premises infrastructure because it is not always appropriate for all systems to be moved in the same way or even at the same time. In addition, NDH currently runs a wide variety of applications in its data centers, from application servers for production applications to databases for testing and development, networking infrastructure, source control software, artificial intelligence systems running on GPUs, and medical platforms for their customers.

Step one is to do an assessment of all the systems that already exist. This will include servers, databases, apps, where they are stored, network connections, and system dependencies. For instance, there could be a dependency between a customer-facing application and another system such as a specific database, internal API, authentication system, or file storage facility. If these system dependencies were not established before migration, there might be service outages for NDH.

3.2 Phased Migration Approach

Rather than migrating all of the systems onto AWS in one single effort, NDH needs to adopt a step-by-step migration approach. This will reduce risks, give the technical staff an opportunity to gain expertise in using AWS, and ensure that each phase is thoroughly

checked before moving on to the next one.

This stage should focus on planning and laying down the basic foundation. At this stage, NDH will need to lay down the AWS account structure, networking, access control, logging requirements, and baseline security. This will involve creating the basic AWS architecture, designing the Virtual Private Cloud (VPC), setting public and private subnets, implementing identity and access controls, and putting in place logging and auditing capabilities. The importance of establishing these basic foundations before migrating is that they ensure the migration of the workload takes place in a secure environment.

Phase two should see the transfer of lower-risk workloads. The development and testing environments should be considered as good options for initial migration due to the significance they hold for NDH operations and also their safety compared to the critical nature of production environments. Such migration will provide NDH with a chance to get used to AWS services without posing any risks to its production applications.

The last phase will see the migration of production workloads. Production workloads should only be transferred after successful testing, having a reliable backup strategy, proper rollback strategies and ensuring that appropriate security protocols are in place. Applications should be deployed to various availability zones in order to achieve better resilience. Data should be backed up before migration, and performance testing should also be conducted.

3.3 Re-hosting, Re-platforming and Refactoring

There is a need for NDH to use different migration techniques depending on the workload. This is because there are different types of workloads such as development systems, production, database, storage, and artificial intelligence.

Re-hosting, also referred to as lift-and-shift, entails moving applications or servers into AWS without much modification. Re-hosting would be applicable in instances where there is a legacy internal software system or when the migration needs to be done quickly. In one such instance, the existing application server may be moved to Amazon EC2 without any modifications being made to the application itself. The re-hosting approach can be viewed as being quick and more straightforward compared to redeveloping an entire software application.

Re-platforming involves some adjustments to make sure that the application workload can take advantage of managed AWS services. Instead of running the database on a self-managed server, NDH can utilize Amazon RDS service. It will help eliminate the need for maintaining and patching the database, monitoring, backup and availability of

the database. File storage, backups, logs, and static content can all be hosted by Amazon S3 services instead of being placed on file servers. The advantage of re-platforming is that it allows one to enjoy cloud services without any program change.

Refactoring should be applied to NDH applications that need higher scalability and resilience. Refactoring could prove to be valuable for NDH applications dealing directly with patients or having changing demand. This is because refactoring could involve dividing monolithic applications into smaller parts, improving API integration, or making the application work better with AWS-managed services. However, refactoring takes much more time than other methods and demands higher programming efforts.

4 Proposed AWS Cloud Architecture

4.1 Network Architecture

For NDH, it is recommended that the proposed AWS architecture utilize Amazon Virtual Private Cloud. The use of VPC will enable NDH to create its own secure cloud environment comprising application servers, databases, storage solutions, and security mechanisms. This is because NDH operates within the health care sector where critical systems have to be protected from unnecessary exposure to the public Internet.

The VPC will be designed in such a way that it encompasses at least two Availability Zones. Availability Zones refer to different data centers within an AWS region. Using two Availability Zones makes the application resilient since if one Availability Zone goes down, it is still possible to have the application running in another Availability Zone. This will fulfill NDH's requirement for resiliency while reducing downtime among its healthcare clients.

The network needs to be segregated into public and private subnets. Public subnets need to be used for applications that use the internet traffic; these could be the application load balancers. The private subnets, on the other hand, can have the application servers, databases, and other internal applications. This will give users an opportunity to use NDH's applications through the public gateway.

It is recommended that the security group be employed in managing traffic at each of the architecture's levels. For instance, the Application Load Balancer needs to receive the HTTPS traffic coming from the consumers while the EC2 application server should only be receiving traffic from the load balancer. In addition, the database layer should only receive traffic from the permitted application servers. With this approach, the company will prevent unnecessary access, thereby safeguarding the sensitive information stored.

4.2 Compute and Application Layer

NDH should make use of Amazon EC2 to host their virtual servers. EC2 provides customizable computing power, making it possible for NDH to select various instance types depending on their needs. For instance, common web applications might utilize general-purpose instances, while complex applications may require optimized instances.

For scaling to be easier, it is recommended that the EC2 instances are launched using Auto Scaling groups. With Auto Scaling, the number of instances will be increased depending on demand and reduced if the demand goes down (Amazon Web Services 2026*d*). This is one of the problems that will easily be sorted out because currently, the number of physical servers required cannot increase until new ones are acquired.

Elastic Load Balancing needs to be located in front of the application layer. It will be responsible for distributing the traffic across several EC2 instances that will operate in different Availability Zones. The performance and availability will be increased. If an EC2 instance becomes unhealthy, the traffic may be redirected to other healthy instances. It is crucial for NDH since its healthcare systems need to be reliable for the company's customers and end-users (Amazon Web Services 2026*c*).

4.3 Database and Storage Layer

In the context of a relational database workload, the company should make use of Amazon RDS and Amazon Aurora. The utilization of AWS-managed database service reduces the overhead required when it comes to managing database services, which is traditionally done by database administrators (Amazon Web Services 2026*e*). AWS-managed database service performs several tasks for you, including management of the database server, OS updates, backup, and the failover configuration.

If relevant, Multi-AZ deployment of the production database should be implemented. The purpose of doing so is to make sure that there is a secondary database instance running in a different Availability Zone in case of failure of the primary database instance. The relevance of this approach to NDH is that the production healthcare application requires a high availability environment. The automated backup option should also be utilized in order to prevent data loss.

Objects should be stored using Amazon S3. Examples include application logs, backups, export reports, static files, data sets, and archives. S3 offers unparalleled scalability, enabling NDH to handle its increasing storage needs without investing in physical storage equipment. S3 buckets' access privileges should be managed carefully so that confidential files are not publicly accessible.

It is recommended that S3 Life-cycle policies be put into practice in order to cut down expenses. For instance, documents that are not frequently accessed can be stored in less expensive storage types while important documents such as reports and backups should be saved in normal storage. In doing so, NDH will succeed in achieving its goal of cutting costs.

4.4 Global Delivery and High Availability

NDH services both organizations and individuals in various locations around the world, including Europe, North America, and the Asia-Pacific region. Therefore, the AWS infrastructure must allow access globally and deliver low latency. The Amazon CloudFront product should be leveraged as the content delivery network for storing static content nearer to consumers. This could enhance the efficiency of web-based and mobile-based applications by reducing the distance that users have to travel to access their required data.

The Amazon Route 53 product can be employed to manage DNS and direct traffic to the right endpoints. Route 53 can direct customers to the best application endpoint and ensure failover in case one service fails. This will benefit NDH as the organization must provide reliable access to applications in different geographic locations.

5 Security and Compliance Strategy

5.1 Identity and Access Management

Security is a high priority at NDH since it deals with the health care industry, and the workforce is mainly working remotely. Therefore, when transitioning to AWS, a security by design strategy must be employed, where access controls and security systems are integrated from the beginning into the cloud infrastructure.

In order to restrict access to the AWS service, you will need to employ AWS Identity and Access Management (IAM). The least privilege approach applies to the NDH, where each user or service should only be provided with privileges required to accomplish a particular task. For instance, developers require access to the development environment; however, they should not be immediately provided with access to the production database and security configurations. On the other hand, compliance requires access to the audit log files; yet, they should not alter the application's infrastructure.

NDH must also split the responsibilities among various teams. Administrators, developers, testers, IT support staff, and security team members should each have defined responsibil-

ities. This reduces chances for over permissions and creates more responsibility. Admin accounts should have two-factor authentication. The use of shared accounts should not occur as this makes it hard to identify who performed a specific task.

5.2 Network Security

The AWS Virtual Private Cloud architecture needs to have some considerations for network security. NDH must not connect critical systems directly to the internet. Public subnets must contain resources that need internet connectivity, such as the Application Load Balancer. The application server, database, and other internal applications must be placed in private subnets.

Security groups must act as virtual firewalls for controlling communications between resources. For instance, the Application Load Balancer must allow communication using HTTPS protocol from clients, while EC2 instances must only allow communication from the Application Load Balancer (Amazon Web Services 2026*b*). Databases must only receive requests from authorized application servers. The above approach will ensure unnecessary communication does not occur and the impact of any breach of system security is minimized.

It would be necessary to employ AWS WAF to protect NDH's applications from web-based threats. This is essential because NDH operates in a highly data-sensitive environment and offers both web and mobile applications for its clients. AWS Shield can be employed to guard against potential DDoS attacks.

5.3 Data Protection

Data security is necessary since NDH works with healthcare data. It means that the data should be secured during storage and transportation. All the stored data in Amazon web services such as Amazon RDS, Amazon S3, and backups should be encrypted. Encryption keys can be generated and controlled by using AWS KMS.

It is also necessary to encrypt the data transferred from users, apps, and backend services. The usage of HTTPS and TLS will help to decrease the risk of interception of sensitive data. Secure connections can also be applied when transferring healthcare and authentication data between services internally.

Information in Amazon S3 needs to be safeguarded through proper bucket policies and access management. Public access must be minimized unless there is a valid reason for public access. Sensitive information, AI training information, reports, backup copies, and logs must only be accessible to authorized individuals and applications. Finally, NDH

must develop its data retention policy to minimize unnecessary storage.

6 Cost Optimization and Operational Management

6.1 Cost Control

Cost optimization is an integral part of NDH’s migration strategy since one of the key drawbacks of the current on-premises setup is its high cost. The on-premises solution requires substantial upfront investment in servers, storage systems, network equipment, GPUs, electricity, cooling systems, and other hardware-related costs. Transitioning to AWS allows NDH to shift from CAPEX to OPEX, allocating funds for resources depending on usage.

Costs associated with Amazon EC2 should be controlled by implementing right-sizing and Auto Scaling. Right-sizing involves choosing the most appropriate instance type to match the demands of each activity. In other words, testing and development machines do not need to consume the same amount of computing resources as production environments unless otherwise required. Auto Scaling may also be used to prevent waste by increasing compute resources during busy times and decreasing them during idle periods.

The cost of storage should not be overlooked either. Using lifecycle policies on Amazon S3 would allow moving some of the backups, logs, and archives to a cheaper tier of storage (Amazon Web Services 2026f). The frequently accessed files would remain in the usual storage, whereas the infrequently accessed ones would be moved to less costly storage types. This is critical for NDH due to the large amount of logging, reports, backups, and analytics data that can build up over time.

6.2 Backup and Disaster Recovery

Backup and disaster recovery strategies need to be included in the overall AWS migration plan for NDH from the very beginning since NDH offers services relating to health care. The failure to achieve this may lead to adverse effects on the business and its reputation because of the sensitive nature of the data involved.

Recovery time and recovery point objectives need to be established by NDH for their critical applications. Recovery time objectives define the required duration within which the system needs to be restored after failure while recovery point objectives indicate the acceptable amount of data loss. This information will help determine the frequency of backups and resiliency of workloads.

Amazon RDS automated backups must be implemented in production databases. If feasible, Multi-AZ architecture should be employed to guarantee database accessibility, despite one Availability Zone becoming unavailable. Data versioning must be configured for important data stored within Amazon S3 to avoid accidental deletion and modification.

7 Conclusion

The recommended AWS architecture fulfills the needs of the business of NDH through the use of services such as Amazon VPC, EC2, Auto Scaling, Elastic Load Balancing, RDS, S3, CloudFront, and Route 53 (Amazon Web Services 2026a). Such services provide a scalable and highly available platform that can support healthcare applications used by customers, remote development teams, a global audience, and even AI. On top of that, the issue of security cannot be overlooked when designing an IT architecture for any business. AWS provides several tools to support that effort.

In general, the AWS infrastructure architecture is more flexible and modern compared to that of the on-premises solution currently used by NDH. In the event of careful planning, proper management, and continuous optimization post-implementation, NDH will be able to offer a cloud-based environment that enables scalability, increased service reliability, better security, and reduced need for managing hardware.

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